Learning to Sense: Uncertainty-Aware Active Sensing for Ocean Physical Fields via Bayesian Tensor Completion

Zhongtao Chen
Department of EEE
The University of Hong Kong
ztchen@eee.hku.hk

Bingyang Cheng
Department of EEE
The University of Hong Kong
bycheng@eee.hku.hk

Wenqiang Pu
Shenzhen Research Institute of Big Data
The Chinese University of Hong Kong, Shenzhen
wenqiangpu@cuhk.edu.cn

Lei Cheng
ISEE College
Zhejiang University
lei_cheng@zju.edu.cn

Yik-Chung Wu
Department of EEE
The University of Hong Kong
ycwu@eee.hku.hk

***Abstract*—Reconstructing physical fields from sparse measurements is a fundamental problem in scientific exploration. Most existing approaches treat measurement acquisition and field reconstruction as separate tasks. In contrast, modern sensing platforms enable adaptive measurement design, motivating active sensing strategies that sequentially select informative samples based on the current field estimate. In this paper, an uncertainty-aware active sensing framework is developed in which new measurements are selected by maximizing the posterior predictive variance. To obtain the posterior distribution on physical fields defined on multi-dimensional spatial grids, a Bayesian tensor completion formulation is adopted. However, exact posterior inference in this model is intractable, rendering direct evaluation of the posterior predictive variance infeasible. To address this challenge, an approximation of the posterior predictive variance based on Gibbs sampling is derived, and a theoretical convergence guarantee is established. Experimental results on both synthetic and real-world datasets demonstrate that the proposed active sensing method outperforms uniform random sampling, achieving more accurate field reconstruction with fewer measurements.**

***Index Terms*—Active sensing, physical field reconstruction, tensor completion, Bayesian inference**

I. Introduction

Physical fields encode the fundamental structure and dynamics of natural environments, governing phenomena across the oceans, atmosphere, and solid Earth—from marine acoustic patterns to atmospheric pressure gradients and stresses embedded in the Earth's crust. Owing to practical constraints such as high deployment cost, limited energy budgets, and restricted environmental accessibility, measurements of physical fields are typically sparse [1]–[5]. Reconstructing the underlying field from such incomplete observations therefore constitutes a central challenge in scientific exploration.

While a wide range of reconstruction methods have been developed to address this challenge [1]–[5], most existing approaches assume that measurement locations are fixed a priori and independent of the reconstruction process. In many modern sensing platforms, however, mobile or reconfigurable sensors are able to adaptively decide where to sample next. This capability motivates *active sensing*, in which measurement acquisition is guided by previously collected data to improve reconstruction accuracy using fewer measurements.

An essential question in active sensing is how to assess the potential value of a candidate measurement location. Uncertainty-aware strategies address this question by exploiting the uncertainty associated with the current reconstruction. The underlying rationale is that measurements acquired in regions of high uncertainty are more likely to reduce ambiguity in the reconstructed field than passive sampling schemes that do not leverage information from the current field estimate [6].

Implementing uncertainty-aware active sensing, however, requires a principled mechanism for uncertainty quantification. The Bayesian framework provides a natural solution through the posterior distribution, which characterizes uncertainty in the reconstructed field conditioned on observed data. To effectively exploit posterior uncertainty, this framework must be paired with a suitable Bayesian model of the underlying physical field. In many ocean sensing applications, physical field observations are collected on multi-dimensional spatial grids, naturally giving rise to tensor-structured data. Bayesian tensor models [7]–[13] provide a faithful modeling for such data. Unfortunately, exact posterior inference in Bayesian tensor models is generally intractable [7]–[13], which prevents direct evaluation of posterior uncertainty measures required for uncertainty-aware active sensing.

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In this paper, we develop an uncertainty-aware active sensing framework based on Bayesian tensor completion. The posterior variance of the unobserved entry is employed as the acquisition criterion to guide adaptive measurement selection. To obtain the posterior distribution of the physical field, we adopt the Bayesian Tucker completion (BTC) model [7] as an illustrative example. To address the intractability of the exact posterior variance computation, we derive a Gibbs sampling–based approximation and establish the theoretical convergence guarantee. Experimental results on both synthetic and real-world datasets demonstrate that the proposed framework outperforms uniform random selection in terms of reconstruction accuracy.

The remainder of this paper is organized as follows. Section II introduces the uncertainty-aware active sensing framework. Section III presents the Gibbs sampling-based approximation and guarantee. Numerical results demonstrating the effectiveness of the proposed method are provided in Section IV, and conclusions are drawn in Section V.

Notation: Scalars, vectors, matrices, and tensors are denoted by x, $\boldsymbol{x}$, $\mathbf{X}$, and $\boldsymbol{\mathcal{X}}$, respectively. An $N \times N$ diagonal matrix with diagonal entries $a_1, \cdots, a_N$ is denoted by $\operatorname{diag}\{a_1, \cdots, a_N\}$. The superscript $^\top$ denotes the transpose, $\operatorname{vec}(\cdot)$ the vectorization operator, and $\otimes$ the Kronecker product. Expectation and variance are denoted by $\mathbb{E}[\cdot]$ and $\mathbb{V}[\cdot]$, respectively. The multivariate normal probability density function (pdf) of a vector $\boldsymbol{x}$ with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ is denoted by $\mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$, and the Gamma pdf of a scalar x with shape parameter a and rate parameter b is denoted by $\operatorname{Ga}(x|a, b)$. For an index set Ω, $|\Omega|$ denotes its cardinality.

II. Uncertainty-Aware Active Sensing

In certain physical field reconstruction tasks, the sensing platform is capable of adaptively selecting measurement locations. For example, in ocean sensing applications, a ship can be maneuvered to different locations to collect new observations. This capability naturally raises the question of which location should be sensed next. Although uniform random selection is commonly adopted, it does not exploit information from previously collected observations and may therefore lead to suboptimal reconstruction performance.

A more informative sensing strategy can be developed by explicitly exploiting model uncertainty. The underlying rationale is that acquiring measurements in regions of high predictive uncertainty is more likely to reduce model ambiguity and improve reconstruction accuracy than uniform sampling [6]. Specifically, given the observations $\boldsymbol{\mathcal{Y}}_\Omega$ collected on the index set Ω, an uncertainty-aware acquisition strategy is adopted that selects the next sensing location by maximizing the posterior predictive variance [6]:

$$
\begin{aligned}
\max_{\boldsymbol{i} \notin \Omega} \quad & \mathbb{V}_{p(\boldsymbol{\Theta}|\boldsymbol{\mathcal{Y}}_\Omega)}\left[\boldsymbol{\mathcal{Y}}_{\boldsymbol{i}}|\boldsymbol{\mathcal{Y}}_\Omega\right] \\
& = \mathbb{E}_{p(\boldsymbol{\Theta}|\boldsymbol{\mathcal{Y}}_\Omega)}\left[\boldsymbol{\mathcal{Y}}_{\boldsymbol{i}}^2\right] - \left(\mathbb{E}_{p(\boldsymbol{\Theta}|\boldsymbol{\mathcal{Y}}_\Omega)}\left[\boldsymbol{\mathcal{Y}}_{\boldsymbol{i}}\right]\right)^2, \qquad (1)
\end{aligned}
$$

where the posterior distribution of the model parameters $\boldsymbol{\Theta}$ is given by

$$
p(\boldsymbol{\Theta}|\boldsymbol{\mathcal{Y}}_\Omega) = \frac{p(\boldsymbol{\mathcal{Y}}_\Omega|\boldsymbol{\Theta})p(\boldsymbol{\Theta})}{\int p(\boldsymbol{\mathcal{Y}}_\Omega|\boldsymbol{\Theta})p(\boldsymbol{\Theta})d\boldsymbol{\Theta}}. \qquad (2)
$$

Under the additive white Gaussian noise (AWGN) assumption, problem (1) admits a simplified form. Specifically, assume that each observation satisfies $\boldsymbol{\mathcal{Y}}_{\boldsymbol{i}} = \boldsymbol{\mathcal{X}}_{\boldsymbol{i}}(\boldsymbol{\Theta}) + \boldsymbol{\mathcal{E}}_{\boldsymbol{i}}$, where the latent field $\boldsymbol{\mathcal{X}}_{\boldsymbol{i}}(\boldsymbol{\Theta})$ depends on the model parameters $\boldsymbol{\Theta}$, and the noise term $\boldsymbol{\mathcal{E}}_{\boldsymbol{i}}$ is independently drawn from a Gaussian distribution $\mathcal{N}(\boldsymbol{\mathcal{E}}_{\boldsymbol{i}}|0, \tau^{-1})$ with zero mean and precision τ. The posterior predictive variance can then be decomposed as

$$
\begin{aligned}
&\mathbb{E}_{p(\boldsymbol{\Theta}|\boldsymbol{\mathcal{Y}}_\Omega)}\left[(\boldsymbol{\mathcal{X}}_{\boldsymbol{i}}(\boldsymbol{\Theta}) + \boldsymbol{\mathcal{E}}_{\boldsymbol{i}})^2\right] \\
&\qquad - \left(\mathbb{E}_{p(\boldsymbol{\Theta}|\boldsymbol{\mathcal{Y}}_\Omega)}\left[\boldsymbol{\mathcal{X}}_{\boldsymbol{i}}(\boldsymbol{\Theta})\right] + \mathbb{E}_{p(\boldsymbol{\Theta}|\boldsymbol{\mathcal{Y}}_\Omega)}\left[\boldsymbol{\mathcal{E}}_{\boldsymbol{i}}\right]\right)^2 \\
&= \mathbb{E}_{p(\boldsymbol{\Theta}|\boldsymbol{\mathcal{Y}}_\Omega)}\left[\boldsymbol{\mathcal{X}}_{\boldsymbol{i}}^2(\boldsymbol{\Theta})\right] - \left(\mathbb{E}_{p(\boldsymbol{\Theta}|\boldsymbol{\mathcal{Y}}_\Omega)}\left[\boldsymbol{\mathcal{X}}_{\boldsymbol{i}}(\boldsymbol{\Theta})\right]\right)^2 + \mathbb{E}_{p(\boldsymbol{\Theta}|\boldsymbol{\mathcal{Y}}_\Omega)}\left[\tau^{-1}\right]. \qquad (3)
\end{aligned}
$$

Since the expected noise variance does not depend on the index $\boldsymbol{i}$, problem (1) is equivalent to maximizing the posterior variance of the latent field:

$$
\max_{\boldsymbol{i} \notin \Omega} \; \mathbb{E}_{p(\boldsymbol{\Theta}|\boldsymbol{\mathcal{Y}}_\Omega)}\left[\boldsymbol{\mathcal{X}}_{\boldsymbol{i}}^2(\boldsymbol{\Theta})\right] - \left(\mathbb{E}_{p(\boldsymbol{\Theta}|\boldsymbol{\mathcal{Y}}_\Omega)}\left[\boldsymbol{\mathcal{X}}_{\boldsymbol{i}}(\boldsymbol{\Theta})\right]\right)^2. \qquad (4)
$$

III. Active Sensing via Bayesian Tensor Completion

Since the active sensing problem in (4) requires evaluation of the posterior variance of the latent field, a Bayesian model for the underlying physical field is necessary. To represent the multi-dimensional structure of the ocean physical field, a Bayesian tensor completion model is adopted. Owing to the analytical intractability of exact posterior inference under this model, a Gibbs sampling–based approximation is developed, together with theoretical convergence guarantees.

As indicated by (4), the acquisition criterion depends on the posterior distribution in (2), necessitating specification of both the likelihood function and the prior distribution. While the likelihood $p(\boldsymbol{\mathcal{Y}}_\Omega|\boldsymbol{\Theta})$ follows from the AWGN assumption, the prior distribution $p(\boldsymbol{\Theta})$ must be carefully designed to capture the structural properties of the underlying physical field. In many ocean sensing applications, the physical field is discretized on a regular grid over a three-dimensional spatial domain, yielding a latent tensor $\boldsymbol{\mathcal{X}}(\boldsymbol{\Theta}) \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, where each entry $\boldsymbol{\mathcal{X}}_{\boldsymbol{i}}(\boldsymbol{\Theta})$ represents the field value at location $\boldsymbol{i} = (i_1, i_2, i_3)$. To faithfully model such tensor-structured data, we adopt the prior distribution proposed in the Bayesian Tucker completion (BTC) model [7] as a representative example. Under the Tucker parameterization, each entry of the latent tensor is expressed as

$$
\boldsymbol{\mathcal{X}}_{\boldsymbol{i}}(\boldsymbol{\Theta}) = \sum_{r_1=1}^{R_1} \sum_{r_2=1}^{R_2} \sum_{r_3=1}^{R_3} \boldsymbol{\mathcal{G}}_{r_1,r_2,r_3} \prod_{n=1}^{3} \mathbf{U}^{(n)}_{i_n,r_n}. \qquad (5)
$$

where the model parameters $\boldsymbol{\Theta}$ consist of the core tensor $\boldsymbol{\mathcal{G}} \in \mathbb{R}^{R_1 \times R_2 \times R_3}$ and the factor matrices $\mathbf{U}^{(n)} \in \mathbb{R}^{I_n \times R_n}$, and (R_1, R_2, R_3) denotes the multilinear rank. Detailed specifications of the prior distribution over the model parameters $\boldsymbol{\Theta}$ are provided in [7].

Under the specified likelihood and prior distributions, the exact computation of the posterior distribution in (2) is intractable due to the complex dependencies among the model parameters. This intractability, in turn, prevents direct evaluation of the uncertainty-aware active sensing criterion in (4). In the following, we derive an approximation of the active sensing criterion using a sampling-based inference method, namely Gibbs sampling, and establish theoretical guarantees for the resulting approximation.

A. Gibbs Sampling for Active Sensing and Guarantee

When posterior samples are available, the active sensing criterion in (4) can be approximated using the empirical variance of the predictive samples. Specifically, given T samples $\{\boldsymbol{\Theta}^{(t)}\}_{t=1}^{T}$, the criteria in (4) can be approximated as

$$\frac{1}{T}\sum_{t=1}^{T}\big(\boldsymbol{\mathcal{X}}_{\boldsymbol{i}}(\boldsymbol{\Theta}^{(t)})\big)^2 - \Big(\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{\mathcal{X}}_{\boldsymbol{i}}(\boldsymbol{\Theta}^{(t)})\Big)^2. \tag{6}$$

To generate posterior samples for the BTC model [7], we employ Gibbs sampling [14]. This choice is motivated not only by the availability of closed-form full conditional distributions (as will be derived in the next subsection), but also by the theoretical convergence guarantees offered by Gibbs sampling.

In Gibbs sampling, posterior samples are obtained by iteratively drawing each parameter block from its full conditional distribution conditioned on the current values of all other blocks. Specifically, at iteration t, the j-th parameter block is updated according to

$$\boldsymbol{\Theta}_j^{(t)} \sim p\Big(\boldsymbol{\Theta}_j|\{\boldsymbol{\Theta}_k^{(t)}\}_{k<j}, \{\boldsymbol{\Theta}_k^{(t-1)}\}_{k>j}, \boldsymbol{\mathcal{Y}}\Big), \tag{7}$$

where $\boldsymbol{\Theta}_j$ denotes the j-th parameter block in $\boldsymbol{\Theta}$.

Substituting the samples generated by (7) into (6) yields a Monte Carlo estimator of the active sensing criterion. The validity of this estimator is guaranteed by Proposition 1. The proof follows from the ergodic theorem for Markov chains, which states that sample averages of integrable functions along an ergodic Markov chain converge almost surely to their expectations under the invariant distribution [14]. As will be shown in the next subsection, all full conditional distributions of the BTC model [7] are well defined and everywhere strictly positive on a connected continuous parameter space. Consequently, the conditions of Proposition 1 are satisfied.

Proposition 1. *Assume that all full conditional distributions of the Gibbs sampler are well defined and everywhere strictly positive on a connected continuous parameter space. Then, the Monte Carlo estimator in* (6) *converges almost surely to the true value of the active sensing criterion in* (4) *as the number of samples* $T \to \infty$.

B. Deriving Full Conditional Distributions

Implementing the update in (7) requires the full conditional distributions for each parameter block. For brevity, we omit the derivations and directly present the resulting full conditional distributions below.

1) Sampling of $\boldsymbol{\mathcal{G}}$: The full conditional of the core tensor $\boldsymbol{\mathcal{G}}$ is a Gaussian distribution over its vectorization:

$$p(\boldsymbol{\mathcal{G}}|-) = \mathcal{N}\Big(\mathrm{vec}(\boldsymbol{\mathcal{G}})|\boldsymbol{\mu}_{\boldsymbol{\mathcal{G}}}, \boldsymbol{\Sigma}_{\boldsymbol{\mathcal{G}}}\Big), \tag{8}$$

where the covariance and mean are

$$\boldsymbol{\Sigma}_{\boldsymbol{\mathcal{G}}} = \left(\beta\bigotimes_{n=1}^{3}\left\{\lambda_1^{(n)}, \cdots, \lambda_{R_n}^{(n)}\right\} + \tau\sum_{\boldsymbol{i}\in\Omega}\bigotimes_{n=1}^{3}\mathbf{U}_{i_n,:}^{(n)}[\mathbf{U}_{i_n,:}^{(n)}]^{\top}\right)^{-1}, \tag{9}$$

$$\boldsymbol{\mu}_{\boldsymbol{\mathcal{G}}} = \tau\boldsymbol{\Sigma}_{\boldsymbol{\mathcal{G}}}\sum_{\boldsymbol{i}\in\Omega}\boldsymbol{\mathcal{Y}}_{\boldsymbol{i}}\bigotimes_{n=1}^{3}\mathbf{U}_{i_n,:}^{(n)}, \tag{10}$$

where the Kronecker product is defined as $\bigotimes_{n=1}^{3}\mathbf{A}^{(n)} = \mathbf{A}^{(3)}\otimes\mathbf{A}^{(2)}\otimes\mathbf{A}^{(1)}$

2) Sampling of $\mathbf{U}_{i_n,:}^{(n)}, \forall i_n$: For each row factor $\mathbf{U}_{i_n,:}^{(n)}$, the full conditional distribution admits the Gaussian form:

$$p(\mathbf{U}_{i_n,:}^{(n)}|-) = \mathcal{N}\Big(\mathbf{U}_{i_n,:}^{(n)}|\boldsymbol{\mu}_{\mathbf{U}_{i_n,:}^{(n)}}, \boldsymbol{\Sigma}_{\mathbf{U}_{i_n,:}^{(n)}}\Big), \tag{11}$$

where the covariance and mean are

$$\boldsymbol{\Sigma}_{\mathbf{U}_{i_n,:}^{(n)}} = \Bigg(\mathrm{diag}\left\{\lambda_1^{(n)}, \cdots, \lambda_{R_n}^{(n)}\right\} + \tau\boldsymbol{\mathcal{G}}_{(n)}\Big(\sum_{\boldsymbol{i}\in\Omega}\bigotimes_{k\neq n}\mathbf{U}_{i_k,:}^{(n)}[\mathbf{U}_{i_k,:}^{(n)}]^{\top}\Big)\boldsymbol{\mathcal{G}}_{(n)}^{\top}\Bigg)^{-1}, \tag{12}$$

$$\boldsymbol{\mu}_{\mathbf{U}_{i_n,:}^{(n)}} = \tau\boldsymbol{\Sigma}_{\mathbf{U}_{i_n,:}^{(n)}}\boldsymbol{\mathcal{G}}_{(n)}\sum_{\boldsymbol{i}\in\Omega}\Big(\boldsymbol{\mathcal{Y}}_{\boldsymbol{i}}\bigotimes_{k\neq n}\mathbf{U}_{i_k,:}^{(n)}\Big), \tag{13}$$

where $\boldsymbol{\mathcal{G}}_{(n)}$ represents the mode-n unfolding of the core tensor $\boldsymbol{\mathcal{G}}$.

3) Sampling of $\lambda_{r_n}^{(n)}, \forall n, r_n$: The full conditional distribution associated with $\lambda_{r_n}^{(n)}$ is a Gamma distribution:

$$p(\lambda_{r_n}^{(n)}|-) = \mathrm{Ga}\Big(\lambda_{r_n}^{(n)}|a_{\lambda_{r_n}^{(n)}}, b_{\lambda_{r_n}^{(n)}}\Big), \tag{14}$$

where the parameters are

$$a_{\lambda_{r_n}^{(n)}} = a_\lambda^0 + \frac{1}{2}I_n + \frac{1}{2}\prod_{k\neq n}R_k, \tag{15}$$

$$b_{\lambda_{r_n}^{(n)}} = b_\lambda^0 + \frac{1}{2}[\mathbf{U}_{:,r_n}^{(n)}]^{\top}\mathbf{U}_{:,r_n}^{(n)} + \frac{1}{2}\beta\sum_{r_1,\cdots,r_{n-1},r_{n+1},\cdots,r_N}\boldsymbol{\mathcal{G}}_{r_1,r_2,r_3}^2\prod_{k\neq n}\lambda_{r_k}^{(k)}. \tag{16}$$

4) Sampling of β: The full conditional distribution of β takes the form of a Gamma distribution:

$$p(\beta|-) = \mathrm{Ga}\big(\beta|a_\beta, b_\beta\big), \tag{17}$$

where the parameters are

$$a_\beta = a_\beta^0 + \frac{1}{2}\prod_{n=1}^{3}R_n, \tag{18}$$

$$b_\beta = b_\beta^0 + \frac{1}{2}\sum_{r_1,r_2,r_3}\boldsymbol{\mathcal{G}}_{r_1,r_2,r_3}^2\prod_{n=1}^{3}\lambda_{r_n}^{(n)}. \tag{19}$$

5) Sampling of τ: Finally, the full conditional for the noise precision τ is given by a Gamma distribution:

$$p(\tau|-) = \mathrm{Ga}\big(\tau|a_\tau, b_\tau\big), \tag{20}$$

where the parameters are

$$a_\tau = a_\tau^0 + \frac{1}{2}|\Omega|, \tag{21}$$

$$b_\tau = b_\tau^0 + \frac{1}{2}\sum_{\boldsymbol{i}\in\Omega}\big(\boldsymbol{\mathcal{Y}}_{\boldsymbol{i}} - \boldsymbol{\mathcal{X}}_{\boldsymbol{i}}\big)^2. \tag{22}$$

To summarize, the Gibbs sampling algorithm iteratively samples each parameter block according to the full conditional distributions from (8) to (20). The initialization of the sampler follows the same setup as in [7]. The overall computational complexity of the Gibbs sampler is $O(T\sum_{n=1}^{3} I_n R_n^3)$.

IV. Numerical Results

To validate the effectiveness of the proposed method, we first conduct experiments on synthetic data. The synthetic tensors are of size $10\times10\times10$ with multilinear rank $(2,2,2)$. The entries of the core tensor and factor matrices are independently drawn from the standard normal distribution and assembled according to the Tucker model in (5). The noisy observation tensor is generated as $\boldsymbol{\mathcal{Y}} = \boldsymbol{\mathcal{X}} + \boldsymbol{\mathcal{N}}$, where the noise tensor is defined as $\boldsymbol{\mathcal{N}} = \sigma\boldsymbol{\mathcal{Z}}$, with each entry of $\boldsymbol{\mathcal{Z}}$ sampled independently from the standard normal distribution. The noise level σ is determined by the prescribed signal-to-noise ratio (SNR) via $\sigma = 10^{-\mathrm{SNR}/20}\|\boldsymbol{\mathcal{X}}\|_F/\|\boldsymbol{\mathcal{Z}}\|_F$ [13]. The SNR is set to 20 dB for synthetic data. The reported results are averaged over 50 Monte Carlo trials.

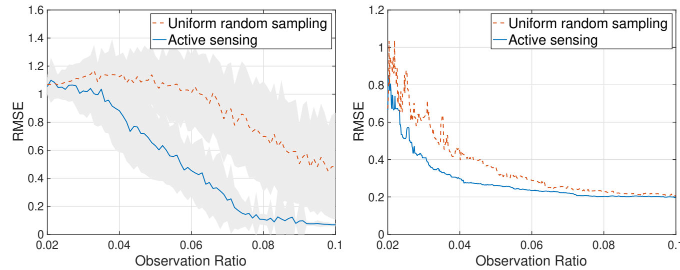
The initial observed entries are selected uniformly at random with an observation ratio of 2%. The sensing process is then carried out for 80 iterations, where one additional entry is acquired at each iteration, resulting in a final observation ratio of 10%. For inference, the Gibbs sampler is initialized with a multilinear rank of $(5,5,5)$ and run for 200 iterations, with the first 100 samples discarded as burn-in.

Reconstruction performance is evaluated using the root mean squared error (RMSE) between the ground-truth tensor $\boldsymbol{\mathcal{X}}$ and the reconstructed tensor $\hat{\boldsymbol{\mathcal{X}}}$, defined as

$$\mathrm{RMSE} = \frac{\left\|\boldsymbol{\mathcal{X}} - \hat{\boldsymbol{\mathcal{X}}}\right\|_F}{\|\boldsymbol{\mathcal{X}}\|_F}, \tag{23}$$

In Fig. 1a, the lines represent the mean RMSE, while the shaded regions indicate one standard deviation across trials. The proposed uncertainty-aware active sensing method consistently achieves lower RMSE than uniform random sampling across all observation ratios. Moreover, as the observation ratio increases, the variance of the proposed method steadily decreases, whereas the variance under uniform random sampling remains comparatively high, indicating improved reliability of the proposed approach.

We next verify the effectiveness of the proposed method on a real-world South China Sea sound speed field (SSF) dataset [5]. The dataset covers a spatial region of $152\mathrm{km}\times152\mathrm{km}\times$



(a) Synthetic data (b) South China Sea SSF data

Fig. 1: Performance of reconstruction accuracy.

$190\mathrm{m}$, with a horizontal resolution of $8\mathrm{km}$ and a vertical resolution of $10\mathrm{m}$, resulting in a tensor of size $20\times20\times20$. The observation tensor is generated by adding Gaussian noise to the ground-truth tensor, with the SNR set to 20 dB. The sensing process is initialized with an observation ratio of 2% and proceeds for 640 iterations, yielding a final observation ratio of 10%. The Gibbs sampler is configured using the same settings as in the synthetic data experiments.

In Fig. 1b, to alleviate abrupt fluctuations, the RMSE curves are smoothed using a median filter with a window size of 7. Consistent with the results on synthetic data, the proposed method consistently outperforms uniform random sampling in terms of RMSE across all observation ratios, demonstrating its effectiveness in real-world ocean sensing applications.

V. Conclusions

In this paper, an uncertainty-aware active sensing framework for ocean physical field reconstruction has been proposed. The active sensing criterion is based the posterior variance of a Bayesian tensor completion model. To address the intractability of exact posterior variance computation, Gibbs sampling-based approximation has been derived with theoretical justification. Experimental results on both synthetic and real-world datasets have demonstrated that the proposed method consistently outperforms uniform random sampling, achieving higher reconstruction accuracy with fewer measurements.

References

[1] S. Li, L. Cheng, X. Fu, and J. Li, "Zero-shot reconstruction of ocean sound speed field tensors: A deep plug-and-play approach," *The Journal of the Acoustical Society of America*, vol. 155, no. 5, pp. 3475–3489, 2024.

[2] S. Li, L. Cheng, J. Li, Z. Wang, and J. Li, "Learning data distribution of three-dimensional ocean sound speed fields via diffusion models," *The Journal of the Acoustical Society of America*, vol. 155, no. 5, pp. 3410–3425, 2024.

[3] C. Gao, L. Cheng, T. Zhang, and J. Li, "Unveiling the spatial-temporal dynamics: Diffusion-based learning of conditional distribution for range-dependent ocean sound speed field forecasting," *The Journal of the Acoustical Society of America*, vol. 156, no. 5, pp. 3554–3573, 2024.

[4] P. Chen, Y. Sun, L. Cheng, Y. Yang, W. Li, Y. Liu, W. Liu, J. Bian, and S. Fang, "Generating full-field evolution of physical dynamics from irregular sparse observations," *arXiv preprint arXiv:2505.09284*, 2025.

[5] P. Chen, S. Li, L. Cheng, X. Fu, Y.-C. Wu, and S. Theodoridis, "Fieldformer: Self-supervised reconstruction of physical fields via tensor attention prior," *IEEE Transactions on Signal Processing*, 2025.

[6] B. Settles, "Active learning literature survey," 2009.

[7] Q. Zhao, L. Zhang, and A. Cichocki, "Bayesian sparse tucker models for dimension reduction and tensor completion," *arXiv preprint arXiv:1505.02343*, 2015.

[8] Z. Long, C. Zhu, J. Liu, and Y. Liu, "Bayesian low rank tensor ring for image recovery," *IEEE Transactions on Image Processing*, vol. 30, pp. 3568–3580, 2021.

[9] L. Cheng, Z. Chen, Q. Shi, Y.-C. Wu, and S. Theodoridis, "Towards flexible sparsity-aware modeling: Automatic tensor rank learning using the generalized hyperbolic prior," *IEEE Transactions on Signal Processing*, vol. 70, pp. 1834–1849, 2022.

[10] X. Tong, L. Cheng, and Y.-C. Wu, "Bayesian tensor tucker completion with a flexible core," *IEEE Transactions on Signal Processing*, vol. 71, pp. 4077–4091, 2023.

[11] L. Xu, L. Cheng, N. Wong, and Y.-C. Wu, "Tensor train factorization under noisy and incomplete data with automatic rank estimation," *Pattern Recognition*, vol. 141, p. 109650, 2023.

[12] ——, "To fold or not to fold: Graph regularized tensor train for visual data completion," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025.

[13] Z. Chen, L. Cheng, Y.-C. Wu, and H. V. Poor, "Rank-revealing Bayesian block-term tensor completion with graph information," *IEEE Transactions on Signal Processing*, vol. 74, pp. 654–669, 2026.

[14] C. P. Robert, G. Casella, and G. Casella, *Monte Carlo statistical methods*. Springer, 1999, vol. 2.